

```

758
*****
*****
758! *****
759
760
*****
*****
760! *****
761
762
*****
*****
762! *****
763 20May09 When subdividing activity codes into pseudo codes to the
CARMM program, these are the
763! places where they also
764 will need to be added:
765 Below where %addActvCodeToMxMail is being
called in (appends the
765! psuedo code to the mix mail map)
766 Below where Proc format is formatting all
the activity codes
767 In the excel file (ActivityCodesMaster
03AUG09.xls)
768 In the sas code of "mapclassRecast.sas"
NOTE not needed for delivery
768! cost model, as not
769 using DMM field.
770 In the mixed mail/direct actv code match file
"MxMailCode.csv" - add new activity
770! codes for Std Mail
771 Be sure to use the correct field in ActivityCodeMaster -
NewCRAProd
772 select KL.*, ac.NewCRAProd as class2
773 FY25: Barcodes -Intelligent Mail Indicia (IMI) replaces
Information Based Indicia
773! (IBI)
774
*****
*****
774! *****
775 REMOVE COMMENT ON SELECT CASING DEFINITION SECTION TO USE FOR CASING
ONLY!!!
776
*****
*****
776! *****;
777
778 resetline;
1 options MPRINT SYMBOLGEN;
2
3 libname IOCS 'D:\ExternalSSD\Folder 19\FY25\SAS\SASLib';
NOTE: Libref IOCS was successfully assigned as follows:

```

```

      Engine:          V9
      Physical Name: D:\ExternalSSD\Folder 19\FY25\SAS\SASLib
4      %let pathProg = D:\ExternalSSD\Folder 19\FY25\SAS\SASInputFiles;
5      %let pathOut = D:\ExternalSSD\Folder 19\FY25\SAS\KLTables;
6      %let FY = 25;
7      %let FYData = 25;
8
9      %macro assignRouteGroup();
10         *** The following statement assigns route code groups
11         *** so that reports can be created separately for
12         *** these groups.;
13
14         *set rgroup for special runs, for delivery cost model (3
values);
15         *Put Letter Route types in RGroup 1, SPR in RGroup 2, 99 in
RGroup 3;
16
17         if '71' <= route <= '83' then RGroup=1;
18         else if '84' <= route <= '98' then RGroup=2;
19         else RGroup=3;
20
21     %mend;
22
23     data b;
24     set IOCS.PRCPubCL25;
25     title1 'FY 20&FY. Qtr A CARMM System';
26     /***** The following variables are updated using Cluster
dataset
26 ! *****/
27     /***** Updated by Dan Zhang on Nov 16, 2020
27 ! *****/
28     dollar = READCOST; /* Cost (in dollar) for each carrier
reading based on cluster
28 ! sampling method */
29     act1st = ' ';
30     act234 = ' ';
31     F9252 = CRAFTGRP; /* Craft Subscript: 5=Full-Time Carrier;
6=Part-Time Carrier */
32     f264 = WEIGHTCAG; /* CAG Group used in the Control Total
Dollar: A-H */
33
/*****
*****
33 ! *****/
34
35
36     */;
37     *** Subdivide activity codes ***;
38
39     ***This section is used for FCSP by Indicia as requested
annually by PRC ***;
40
41     select (Q23E02);
42     when ('E')          indicia = "Stamped ";

```

```

43     when ('F')
44         select (q23E05);
45             when ('A')          indicia = "PVI          ";
46             when ('B')          indicia = "Meter        ";
47             when ('C')          indicia = "IMI          ";
48             otherwise           indicia = "Error        ";
49     end;
50     when ('G')          indicia = "Permit      ";
51     otherwise           indicia = "Other       ";
52 end;
53
54 if f262 = '1060' then do; *FC Single Piece Letters ;
55     if indicia = "Stamped " then F262 = '1061';
56     else
57         if indicia = "Meter  " then F262 = '1062';
58         else
59             if indicia = "PVI    " then F262 = '1063';
60             else
61                 if indicia = "IMI    " then F262 = '1064';
62                 else
63                     if indicia = "Permit  " then F262 = '1066';
64                     else
65                         F262 = '1065';
66 end;
67 if f262 = '2060' then do; *FC Single Piece Flats ;
68     if indicia = "Stamped " then F262 = '2061';
69     else
70         if indicia = "Meter  " then F262 = '2062';
71         else
72             if indicia = "PVI    " then F262 = '2063';
73             else
74                 if indicia = "IMI    " then F262 = '2064';
75                 else
76                     if indicia = "Permit  " then F262 = '2066';
77                     else
78                         F262 = '2065';
79 end;
80
81     ***** END ADDITIONAL CODE FOR FCSP BY INDICIA     *****;
82
83
84     ** CARMM Edits;
85     if f264='K' then f9252='9';
86     if f261 ne '4';
87     if dollar=0 and q01=' ' then delete;
88     else do;
89         if f260 in ('84', '89') then f260='87';
90     end;
91     if f9252 in ('5','6'); **carrier craft;
92 run;

```

NOTE: Character values have been converted to numeric values at the places given by:
 (Line):(Column).

```

      85:32    91:12
NOTE: Variable q01 is uninitialized.
NOTE: There were 150303 observations read from the data set
IOCS.PRCPUBCL25.
NOTE: The data set WORK.B has 102200 observations and 304 variables.
NOTE: DATA statement used (Total process time):
      real time           0.19 seconds
      cpu time            0.09 seconds

SYMBOLGEN: Macro variable PATHPROG resolves to D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles
93
94   *Read in file that maps activity codes into mixed mail codes,
95       and add all new activity codes added above;
96   %include "&pathProg\macMxMail_AMS.sas" / source2; /*GET MORE DETAILS
IN THE LOG FOR THIS
96 ! INCLUDE STATEMENT*/
NOTE: %INCLUDE (level 1) file D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\macMxMail_AMS.sas
      is file D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\macMxMail_AMS.sas.
97   +*Macros to read mixed mail activity code mapping, and insert new
activity codes;
98   +
99   +%macro readMxMail(CSVFile,MxMailMap);
100  +*Read file CSVFile (CSV format) with activity codes that belong to
mixed mail codes;
101  +* Prepare final mixed mail mapping dataset;
102  +Data mxmaill;
103  + infile &CSVfile DLM="," MISSOVER firstobs=2;
104  + format mixcode $4.;
105  + format dircode $4.;
106  + input mixcode $ dircode $;
107  +run;
108  +
109  +Data &MxMailMap; set MxMaill;
110  +run;
111  +%mend;
112  +
113  +%macro addActvCodeToMxMail(MxMailMap, actvMatch, actvNew);
114  +*insert new activity code actvNew with same mixed mail cosdes as
actvMatch;
115  +proc sql;
116  + insert into MxMailMap
117  + select mixcode, &actvNew
118  + from mxmaill
119  + where dircode = &actvMatch;
120  +quit;
121  +%mend;
122  +
123  +
124  +*Inserts activity code subsets that Nancy needs for processing,
these are not official

```

```

124!+activity codes;
125 +*AMS 17Nov17;
126 +%macro addActvCodeToMaster(ActivityCodesTable, ActivityCode,
NewCRAProd);
127 +*insert new activity code actvNew with same mixed mail cosdes as
actvMatch;
128 +proc sql;
129 + insert into ActivityCodesTable
130 + set ActivityCode = &ActivityCode.,
131 +     NewCRAProd = &NewCRAProd.;
132 +quit;
133 +%mend;
134 +
135 +
136 +/*
137 +%macro
addMixedMailShapeToMxMail(MxMailMailMap,mxMailNoShape,shapeCode);
138 +*Insert new mixed mail code with shape info, using first digit of
activity code
139 + as last digit of mixed mail code;
140 +proc sql;
141 + insert into MxMailMap
142 +     select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
143 + from mxMail1
144 + where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
145 +quit;
146 +%mend;*/
147 +%macro addMixedMailShapeToMxMail(MxMailMailMap,mxMailNoShape);
148 +*Insert new mixed mail code with shape info, using first digit of
activity code
149 + as last digit of mixed mail code;
150 +%let shapeCode = 1;
151 +proc sql;
152 + insert into MxMailMap
153 +     select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
154 + from mxMail1
155 + where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
156 +%let shapeCode = 2;
157 +proc sql;
158 + insert into MxMailMap
159 +     select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
160 + from mxMail1
161 + where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
162 +%let shapeCode = 3;
163 +proc sql;
164 + insert into MxMailMap
165 +     select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
166 + from mxMail1
167 + where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
168 +%let shapeCode = 4;

```

```

169 +proc sql;
170 +  insert into MxMailMap
171 +    select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
172 + from mxMail1
173 + where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
174 +quit;
175 +%mend;
176 +
177 +
178 +/*
179 +*test macros;
180 +%let pathData = C:\AraSardar\IOCS\CARMM;
181 +%readMxMail("&pathData\MxMailCodeFY14.csv",MxMailMap)
182 +%addActvCodeToMxMail(MxMailMap,'1060','1061');
183 +%addActvCodeToMxMail(MxMailMap,'1060','1062');
184 +%addActvCodeToMxMail(MxMailMap,'1060','1063');
185 +%addActvCodeToMxMail(MxMailMap,'1310','1311');
186 +%addActvCodeToMxMail(MxMailMap,'1310','1312');
187 +%addActvCodeToMxMail(MxMailMap,'1310','1313');
188 +%addActvCodeToMxMail(MxMailMap,'1310','1314');
189 +%addActvCodeToMxMail(MxMailMap,'1310','1315');
190 +%addActvCodeToMxMail(MxMailMap,'1310','1316');
191 +%addActvCodeToMxMail(MxMailMap,'1310','1317');
192 +%addActvCodeToMxMail(MxMailMap,'1310','1318');
193 +%addActvCodeToMxMail(MxMailMap,'1310','1319');
194 +%addActvCodeToMxMail(MxMailMap,'2310','2311');
195 +%addActvCodeToMxMail(MxMailMap,'2310','2312');
196 +%addActvCodeToMxMail(MxMailMap,'2310','2313');
197 +%addActvCodeToMxMail(MxMailMap,'2310','2314');
198 +%addActvCodeToMxMail(MxMailMap,'2310','2315');
199 +%addActvCodeToMxMail(MxMailMap,'2310','2316');
200 +%addActvCodeToMxMail(MxMailMap,'2310','2317');
201 +%addActvCodeToMxMail(MxMailMap,'2310','2318');
202 +%addActvCodeToMxMail(MxMailMap,'2310','2319');
203 +%addActvCodeToMxMail(MxMailMap,'3310','3311');
204 +%addActvCodeToMxMail(MxMailMap,'3310','3312');
205 +%addActvCodeToMxMail(MxMailMap,'3310','3313');
206 +%addActvCodeToMxMail(MxMailMap,'3310','3314');
207 +%addActvCodeToMxMail(MxMailMap,'3310','3315');
208 +%addActvCodeToMxMail(MxMailMap,'3310','3316');
209 +%addActvCodeToMxMail(MxMailMap,'3310','3317');
210 +%addActvCodeToMxMail(MxMailMap,'3310','3318');
211 +%addActvCodeToMxMail(MxMailMap,'3310','3319');
212 +%addActvCodeToMxMail(MxMailMap,'4310','4311');
213 +%addActvCodeToMxMail(MxMailMap,'4310','4312');
214 +%addActvCodeToMxMail(MxMailMap,'4310','4313');
215 +%addActvCodeToMxMail(MxMailMap,'4310','4314');
216 +%addActvCodeToMxMail(MxMailMap,'4310','4315');
217 +%addActvCodeToMxMail(MxMailMap,'4310','4316');
218 +%addActvCodeToMxMail(MxMailMap,'4310','4317');
219 +%addActvCodeToMxMail(MxMailMap,'4310','4318');
220 +%addActvCodeToMxMail(MxMailMap,'4310','4319');
221 +proc sort data=MxMailMap nodup;

```

```

222 +by mixcode dircode;
223 +run;
224 +*/
225 +
226 +***** End macMxMail;
NOTE: %INCLUDE (level 1) ending.
SYMBOLGEN: Macro variable PATHPROG resolves to D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles
MPRINT(READMXMAIL): *Read file CSVFile (CSV format) with activity codes
that belong to mixed
mail codes;
MPRINT(READMXMAIL): * Prepare final mixed mail mapping dataset;
227 %readMxMail("&pathProg\MxMailCodeFY25.csv",MxMailMap); *for
domestic CRA;
MPRINT(READMXMAIL): Data mxmail1;
SYMBOLGEN: Macro variable CSVFILE resolves to "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv"
MPRINT(READMXMAIL): infile "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv"
DLM="," MISSOVER firstobs=2;
MPRINT(READMXMAIL): format mixcode $4.;
MPRINT(READMXMAIL): format dircode $4.;
MPRINT(READMXMAIL): input mixcode $ dircode $;
MPRINT(READMXMAIL): run;

```

```

NOTE: The infile "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv" is:
      Filename=D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv,
      RECFM=V,LRECL=32767,File Size (bytes)=2624,
      Last Modified=13Nov2025:12:44:54,
      Create Time=17Nov2025:10:11:34

```

```

NOTE: 237 records were read from the infile "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv".
      The minimum record length was 9.
      The maximum record length was 9.
NOTE: The data set WORK.MXMAIL1 has 237 observations and 2 variables.
NOTE: DATA statement used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds

```

```

SYMBOLGEN: Macro variable MXMAILMAP resolves to MxMailMap
MPRINT(READMXMAIL): Data MxMailMap;
MPRINT(READMXMAIL): set MxMail1;
MPRINT(READMXMAIL): run;

```

```

NOTE: There were 237 observations read from the data set WORK.MXMAIL1.
NOTE: The data set WORK.MXMAILMAP has 237 observations and 2 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds

```

```

228
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
229  *add subdivided activity codes to mixed mail mappings );
230
231  %addActvCodeToMxMail(MxMailMap,'1060','1061');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1061'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1061' from mxmaill where
dircode = '1060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

```

```

MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      cpu time           0.00 seconds

```

```

MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
232  %addActvCodeToMxMail(MxMailMap,'1060','1062');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1062'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1062' from mxmaill where
dircode = '1060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

```

```

MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds

```

```

MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
233  %addActvCodeToMxMail(MxMailMap,'1060','1063');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1063'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1063' from mxmaill where
dircode = '1060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

```

```

MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):

```



```
real time          0.01 seconds
cpu time           0.01 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
234  %addActvCodeToMxMail(MxMailMap,'1060','1064');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1064'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1064' from mxmaill where
dircode = '1060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
235  %addActvCodeToMxMail(MxMailMap,'1060','1065');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1065'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1065' from mxmaill where
dircode = '1060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
236  %addActvCodeToMxMail(MxMailMap,'1060','1066');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1066'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1066' from mxmaill where
dircode = '1060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
```

```
real time          0.01 seconds
cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
237 %addActvCodeToMxMail(MxMailMap,'2060','2061');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '2061'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '2060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2061' from mxmaill where
dircode = '2060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
real time          0.01 seconds
cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
238 %addActvCodeToMxMail(MxMailMap,'2060','2062');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '2062'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '2060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2062' from mxmaill where
dircode = '2060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
real time          0.01 seconds
cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
239 %addActvCodeToMxMail(MxMailMap,'2060','2063');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '2063'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '2060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2063' from mxmaill where
dircode = '2060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
```

```
real time          0.00 seconds
cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
240 %addActvCodeToMxMail(MxMailMap,'2060','2064');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '2064'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '2060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2064' from mxmaill where
dircode = '2060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
real time          0.00 seconds
cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
241 %addActvCodeToMxMail(MxMailMap,'2060','2065');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '2065'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '2060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2065' from mxmaill where
dircode = '2060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
real time          0.01 seconds
cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
242 %addActvCodeToMxMail(MxMailMap,'2060','2066');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '2066'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '2060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2066' from mxmaill where
dircode = '2060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
```

```
real time          0.00 seconds
cpu time           0.00 seconds
```

```
243
244
245
246 proc sort data=MxMailMap nodup;
247         by dircode mixcode;
248 run;
```

NOTE: There were 273 observations read from the data set WORK.MXMAILMAP.
NOTE: 0 duplicate observations were deleted.

NOTE: The data set WORK.MXMAILMAP has 273 observations and 2 variables.
NOTE: PROCEDURE SORT used (Total process time):

```
real time          0.01 seconds
cpu time           0.01 seconds
```

```
249
250 data carr; set b;
251     Craft=F9252;
252     Actv=F262;
253     *Actv=F9806; *mcz 10Aug08 use F9806 to distribute Int'l mixed
mail;
254     Route=F260;
255     BF=F261;
256     Fin=F263;
257     Cag=F264;
258     Cagfin=cag !! fin;
259
260     **** SELECT CASING DEFINITION *** ;
261     *To restrict to subset that is casing remove comment from next
line;
262     if Q16F03A in ('A','B','C'); ** General casing (3 options) -- use
for Delivery Cost Model
262! inputs;
263
264     if actv in ('5740', '5745', '5741') then actv = '5750'; /*as per
Audrey Hu, 17Nov23*/
265     if actv <= '0900' and actv not in ('0350','0560') then actgrp=1;
266     else if '1020' <= actv <= '4910' or actv in ('0350','0560')
then actgrp=2;
267     else if '5300' <= actv <= '5750' then actgrp=3;
268     else if '5040' <= actv <= '6720' then actgrp=4;
SYMBOLGEN: Macro variable FY resolves to 25
269     TITLE1 "FY 20&fy. Qtr A CARMM SYSTEM";
270 run;
```

NOTE: There were 102200 observations read from the data set WORK.B.
NOTE: The data set WORK.CARR has 26447 observations and 312 variables.

NOTE: DATA statement used (Total process time):

```
real time          0.06 seconds
cpu time           0.04 seconds
```

```

271
272 proc sql;
273     create table smy as
274     select cag, fin, actv, bf, sum(dollar) as dollar
275     from carr
276     group by cag, fin, actv, bf;
NOTE: Table WORK.SMY created, with 362 rows and 5 columns.

```

```

277 quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.02 seconds
      cpu time           0.00 seconds

```

```

278
279
280 data smyx;
281     array bfx{4} bf1 - bf4;
282     do i=1 to 4;
283         set smy;
284         by cag fin actv bf;
285         if i=1 and bf='2' then i=i+1;
286         if i=1 and bf='3' then i=i+2;
287         if i=1 and bf='5' then i=i+3;
288         if i=2 and bf='3' then i=i+1;
289         if i=2 and bf='5' then i=i+2;
290         if i=3 and bf='5' then i=i+1;
291         bfx{i}=dollar;
292         if last.actv then i=4;
293     end;
294     if bf1=. then bf1=0;
295     if bf2=. then bf2=0;
296     if bf3=. then bf3=0;
297     if bf4=. then bf4=0;
298     total=sum(of bf1-bf4);
299     cagfin=cag !! fin;
300     if actv <= '0900' and actv not in ('0350','0560') then
actgrp=1;
301         else if '1020' <= actv <= '4910' or actv in ('0350','0560')
then actgrp=2;
302         else if '5300' <= actv <= '5750' then actgrp=3;
303         else if '5040' <= actv <= '6720' then actgrp=4;
304         output;
305 run;

```

```

NOTE: There were 362 observations read from the data set WORK.SMY.
NOTE: The data set WORK.SMYX has 362 observations and 13 variables.
NOTE: DATA statement used (Total process time):
      real time          0.03 seconds
      cpu time           0.00 seconds

```

```

306
307
308   proc sort data=smyx;
309       by cagfin actgrp actv;
310   run;

```

NOTE: There were 362 observations read from the data set WORK.SMYX.

NOTE: The data set WORK.SMYX has 362 observations and 13 variables.

NOTE: PROCEDURE SORT used (Total process time):

```

      real time          0.01 seconds
      cpu time           0.01 seconds

```

```

311   *-----;
312   *                                           ;
313   * Dollar tally summary by cag fin route actv bf. This      ;
314   * summary file is used in next step for distributing      ;
315   * mixed-mail.                                           ;
316   *                                           ;
317   *-----;

```

```

318
319   proc sql;
320       create table smy2 as
321       select cag, fin, route, actv, bf, sum(dollar) as dollar
322       from carr
323       group by cag, fin, route, actv, bf;

```

NOTE: Table WORK.SMY2 created, with 1232 rows and 6 columns.

```

324   quit;

```

NOTE: PROCEDURE SQL used (Total process time):

```

      real time          0.03 seconds
      cpu time           0.01 seconds

```

```

325
326
327   data smy2;
328       set smy2;
329
330   *Use macro to assign route group;
331       %assignRouteGroup;
MPRINT(ASSIGNROUTEGROUP):   *** The following statement assigns route
code groups *** so that
reports can be created separately for *** these groups.;
MPRINT(ASSIGNROUTEGROUP):   *set rgroup for special runs, for delivery
cost model (3 values);
MPRINT(ASSIGNROUTEGROUP):   *Put Letter Route types in RGroup 1, SPR in
RGroup 2, 99 in RGroup 3;
MPRINT(ASSIGNROUTEGROUP):   if '71' <= route <= '83' then RGroup=1;
MPRINT(ASSIGNROUTEGROUP):   else if '84' <= route <= '98' then RGroup=2;
MPRINT(ASSIGNROUTEGROUP):   else RGroup=3;
332       cagfin=cag !! fin;
333   run;

```

NOTE: There were 1232 observations read from the data set WORK.SMY2.
NOTE: The data set WORK.SMY2 has 1232 observations and 8 variables.
NOTE: DATA statement used (Total process time):
 real time 0.01 seconds
 cpu time 0.00 seconds

```
334
335   data out;
336       array bff{4} dollar1-dollar4;
337       do i=1 to 4;
338           set smy2;
339           by cag fin route actv bf;
340               if i=1 and bf='2' then i=i+1;
341               if i=1 and bf='3' then i=i+2;
342               if i=1 and bf='5' then i=i+3;
343               if i=2 and bf='3' then i=i+1;
344               if i=2 and bf='5' then i=i+2;
345               if i=3 and bf='5' then i=i+1;
346               bff{i}=dollar;
347               if last.actv then i=4;
348       end;
349       if dollar1=. then dollar1=0;
350       if dollar2=. then dollar2=0;
351       if dollar3=. then dollar3=0;
352       if dollar4=. then dollar4=0;
353       drop i cag fin _type_ _freq_ dollar bf;
354       output;
355   run;
```

WARNING: The variable _type_ in the DROP, KEEP, or RENAME list has never been referenced.

WARNING: The variable _freq_ in the DROP, KEEP, or RENAME list has never been referenced.

NOTE: There were 1232 observations read from the data set WORK.SMY2.
NOTE: The data set WORK.OUT has 1232 observations and 8 variables.
NOTE: DATA statement used (Total process time):
 real time 0.02 seconds
 cpu time 0.00 seconds

```
356
357
358   ****Note: this 2nd part of the program is analogous to the **;
359   ****      old program ALA860C0, which performed distribution **;
360   ****      and produced various summary reports.          **;
361   ****
362   ****Note: this 2nd part of the program is analogous to the **;
363   ****      old program ALA860C0, which performed distribution **;
364   ****      and produced various summary reports.          **;
365   ****
366   data out;
367       set out;
368       if '5300' <= actv <= '5750' then mixmail=1;
369       else if '1000' <= actv <= '5000' or actv in ('0350','0560')
370       then mixmail=0;
```

```
368          dollart=sum(of dollar1 - dollar4);
369 run;
```

NOTE: There were 1232 observations read from the data set WORK.OUT.

NOTE: The data set WORK.OUT has 1232 observations and 10 variables.

NOTE: DATA statement used (Total process time):

```
real time          0.00 seconds
cpu time           0.00 seconds
```

```
370
371      *-----;
372      * The following proc SUMMARY gets the preliminary results for ;
373      * Report 7 This corresponds to the old LIOCATT report ALA860P7;
374      *-----;
375 proc sort data=out out=out7;
376     by rgroup route actv;
377     where mixmail ne 1;
378 run;
```

NOTE: There were 1053 observations read from the data set WORK.OUT.

WHERE mixmail not = 1;

NOTE: The data set WORK.OUT7 has 1053 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

```
real time          0.02 seconds
cpu time           0.00 seconds
```

```
379
380 proc  summary data=out7 noprint;
381     var dollar1 dollar2 dollar3 dollar4;
382     output out=rpt7 sum=;
383     by rgroup route actv;
384     /****uncomment print if report is needed***
385     proc print data=rpt7;
386         id rgroup route ;
387         * sum dollar1 dollar2 dollar3 dollar4;
388         by rgroup route;
389         title2 'Report 7';
390         *****/
391 run;
```

NOTE: There were 1053 observations read from the data set WORK.OUT7.

NOTE: The data set WORK.RPT7 has 263 observations and 9 variables.

NOTE: PROCEDURE SUMMARY used (Total process time):

```
real time          0.02 seconds
cpu time           0.00 seconds
```

```
392
393      *****;
394      * Distribute mixed mail costs to direct mail activity codes. ;
395      *****;
396
```



```

397      *-----;
398      * Read the mixed-mail to direct mail code conversion file.      ;
399      * This file tells which direct mail activity codes to associate;
400      * with each mixed-mail activity code.                          ;
401      *-----;
402      ;
403      Data mxmail;
404      set MxMailMap;
405      run;

```

NOTE: There were 273 observations read from the data set WORK.MXMAILMAP.

NOTE: The data set WORK.MXMAIL has 273 observations and 2 variables.

NOTE: DATA statement used (Total process time):

```

      real time          0.00 seconds
      cpu time           0.00 seconds

```

```

406
407      proc sort data=mxmail; by dircode;
408      run;

```

NOTE: There were 273 observations read from the data set WORK.MXMAIL.

NOTE: The data set WORK.MXMAIL has 273 observations and 2 variables.

NOTE: PROCEDURE SORT used (Total process time):

```

      real time          0.00 seconds
      cpu time           0.01 seconds

```

```

409
410      data dirmail (drop= i mixcode);
411          array dir(8) $mix1 -mix8;
412          do i=1 to 8;
413              set mxmail;
414              by dircode;
415              actv=dircode;
416              dir(i)=mixcode;
417              if last.dircode then i=8;
418          end;
419          output;
420      run;

```

NOTE: There were 273 observations read from the data set WORK.MXMAIL.

NOTE: The data set WORK.DIRMAIL has 97 observations and 10 variables.

NOTE: DATA statement used (Total process time):

```

      real time          0.01 seconds
      cpu time           0.00 seconds

```

```

421      *-----;
422      * Direct mail cost summary by cagfin, route and activity      ;
423      *-----;
424      proc sort data=out out=dist2;
425          by cagfin route actv;
426          where mixmail=0; * Contains only direct mail;

```

427 run;

NOTE: There were 1005 observations read from the data set WORK.OUT.

WHERE mixmail=0;

NOTE: The data set WORK.DIST2 has 1005 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time 0.01 seconds

cpu time 0.00 seconds

428

429 proc summary data=dist2 noprint;

430 var dollar1 dollar2 dollar3 dollar4 dollart;

431 output out=dirdol

432 sum=dollar1d dollar2d dollar3d dollar4d dollartd;

433 by cagfin route actv;

434 title2 'dirdol';

435 * proc print data=dirdol;

436 run;

NOTE: There were 1005 observations read from the data set WORK.DIST2.

NOTE: The data set WORK.DIRDOL has 1005 observations and 10 variables.

NOTE: PROCEDURE SUMMARY used (Total process time):

real time 0.01 seconds

cpu time 0.00 seconds

437

438 *-----;

439 * Merge direct mail cost with the conversion table to associate;

440 * each direct mail activity code with mixed mail code that will;

441 * get distribued cost. ;

442 *-----;

443 proc sort data=dirdol;

444 by actv;

445 run;

NOTE: There were 1005 observations read from the data set WORK.DIRDOL.

NOTE: The data set WORK.DIRDOL has 1005 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time 0.01 seconds

cpu time 0.00 seconds

446 proc sort data=dirmail;

447 by actv;

448 run;

NOTE: There were 97 observations read from the data set WORK.DIRMAIL.

NOTE: The data set WORK.DIRMAIL has 97 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time 0.00 seconds

cpu time 0.00 seconds

```

449
450 data dircode1 (drop=mix1-mix8 i);
451     merge dirdol (in=a)
452           dirmail (in=b keep=actv mix1-mix8);
453     by actv;
454     source=a+2*b;
455     if source=2 then delete; * delete activity codes that
contain
456                                     no cost;
457     array dir(8) mix1-mix8;
458     do i=1 to 8;
459         mixcode=dir(i);
460         if mixcode ne ' ' then output;
461         if mixcode= ' ' then i=8;
462     end;
463     *proc print n;
464 run;

```

NOTE: There were 1005 observations read from the data set WORK.DIRDOL.
NOTE: There were 97 observations read from the data set WORK.DIRMAIL.
NOTE: The data set WORK.DIRCODE1 has 3164 observations and 12 variables.
NOTE: DATA statement used (Total process time):
real time 0.01 seconds
cpu time 0.00 seconds

```

465
466 *-----;
467 * Mixed mail cost summary by cagfin, route and actv ;
468 *-----;
469
470 proc sort data=out out=dist1;
471     by cagfin route actv;
472     where mixmail=1; * Contains only mixed mails;
473 run;

```

NOTE: There were 179 observations read from the data set WORK.OUT.
WHERE mixmail=1;
NOTE: The data set WORK.DIST1 has 179 observations and 10 variables.
NOTE: PROCEDURE SORT used (Total process time):
real time 0.00 seconds
cpu time 0.01 seconds

```

474
475 proc summary data=dist1 noprint;
476     var dollar1 dollar2 dollar3 dollar4 dollart;
477     output out=mxdol
478           sum=dollar1i dollar2i dollar3i dollar4i dollarti;
479     by cagfin route actv;
480     title2 'mxdol';
481 run;

```

NOTE: There were 179 observations read from the data set WORK.DIST1.
NOTE: The data set WORK.MXDOL has 179 observations and 10 variables.
NOTE: PROCEDURE SUMMARY used (Total process time):
 real time 0.00 seconds
 cpu time 0.00 seconds

```
482
483      *-----;
484      * Attach summary mixed mail cost to the direct mail activity  ;
485      * code.                                                         ;
486      *-----;
487
488      data mxdol;
489          set mxdol;
490          rename actv=mixcode;
491          *proc print data=mxdol;
492          run;
```

NOTE: There were 179 observations read from the data set WORK.MXDOL.
NOTE: The data set WORK.MXDOL has 179 observations and 10 variables.
NOTE: DATA statement used (Total process time):
 real time 0.01 seconds
 cpu time 0.00 seconds

```
493
494      proc sort data=dircode1;
495          by cagfin route mixcode;
496          run;
```

NOTE: There were 3164 observations read from the data set WORK.DIRCODE1.
NOTE: The data set WORK.DIRCODE1 has 3164 observations and 12 variables.
NOTE: PROCEDURE SORT used (Total process time):
 real time 0.00 seconds
 cpu time 0.00 seconds

```
497      proc sort data=mxdol;
498          by cagfin route mixcode;
499          run;
```

NOTE: There were 179 observations read from the data set WORK.MXDOL.
NOTE: The data set WORK.MXDOL has 179 observations and 10 variables.
NOTE: PROCEDURE SORT used (Total process time):
 real time 0.00 seconds
 cpu time 0.00 seconds

```
500
501      data final ;
502          merge dircode1 (in=a) mxdol (in=b);
503          by cagfin route mixcode;
504          source=a+2*b;
```

```

505
506      * proc print;
507      * title2 'final';
508      run;

```

NOTE: There were 3164 observations read from the data set WORK.DIRCODE1.
NOTE: There were 179 observations read from the data set WORK.MXDOL.
NOTE: The data set WORK.FINAL has 3164 observations and 17 variables.
NOTE: DATA statement used (Total process time):
real time 0.00 seconds
cpu time 0.01 seconds

```

509
510      *-----;
511      * Calculate the total direct mail cost associated with      ;
512      * each mixed mail code.                                     ;
513      *-----;
514
515      proc sort data=final;
516          by cagfin route mixcode;
517      run;

```

NOTE: There were 3164 observations read from the data set WORK.FINAL.
NOTE: The data set WORK.FINAL has 3164 observations and 17 variables.
NOTE: PROCEDURE SORT used (Total process time):
real time 0.01 seconds
cpu time 0.01 seconds

```

518
519      proc summary data=final noprint;
520          var dollar1d dollar2d dollar3d dollar4d dollartd;
521          output out=total1
522              sum=total1d total2d total3d total4d totaltd;
523          by cagfin route mixcode;
524      run;

```

NOTE: There were 3164 observations read from the data set WORK.FINAL.
NOTE: The data set WORK.TOTAL1 has 410 observations and 10 variables.
NOTE: PROCEDURE SUMMARY used (Total process time):
real time 0.01 seconds
cpu time 0.01 seconds

```

525
526      *-----;
527      * Distribute the mixed mail cost to direct mail activity codes ;
528      *-----;
529      data distfile (drop=_type_ _freq_ source);
530          merge final (in=a) total1 (in=b);
531          by cagfin route mixcode;
532
533          if total1d=0 then ratio1=.;

```

```

534         else ratio1= dollar1d/total1d;
535         dist_1=dollar1i *ratio1;
536
537         if total2d=0 then ratio2=.;
538         else ratio2= dollar2d/total2d;
539         dist_2=dollar2i *ratio2;
540
541         if total3d=0 then ratio3=.;
542         else ratio3= dollar3d/total3d;
543         dist_3=dollar3i *ratio3;
544
545     *** The following statements add the undistributed dollar
546     *** for basic functions 1 - 3 to last basic function;
547
548         if (total1d =0 and dollar1i > 0)
549         then dollar4i=dollar4i +dollar1i;
550         if (total2d =0 and dollar2i > 0)
551         then dollar4i=dollar4i +dollar2i;
552         if (total3d =0 and dollar3i > 0)
553         then dollar4i=dollar4i +dollar3i;
554
555     *** Note that the last basic function distribution is based on
556     *** the total cost (sum of four basic functions cost);
557
558         if totaltd=0 then ratiot=.;
559         else ratiot= dollartd/totaltd;
560         dist_4=dollar4i *ratiot;
561
562     *** The following code assigns the undistributed mixed
563     *** dollars to the dist_0 cagegory;
564
565         if actv=' ' then
566         do;
567             actv=mixcode;
568             dist_0=dollarti;
569         end;
570
571     *Use macro to assign route group;
572     %assignRouteGroup;
MPRINT(ASSIGNROUTEGROUP):    *** The following statement assigns route
code groups *** so that
reports can be created separately for *** these groups.;
MPRINT(ASSIGNROUTEGROUP):    *set rgroup for special runs, for delivery
cost model (3 values);
MPRINT(ASSIGNROUTEGROUP):    *Put Letter Route types in RGroup 1, SPR in
RGroup 2, 99 in RGroup 3;
MPRINT(ASSIGNROUTEGROUP):    if '71' <= route <= '83' then RGroup=1;
MPRINT(ASSIGNROUTEGROUP):    else if '84' <= route <= '98' then RGroup=2;
MPRINT(ASSIGNROUTEGROUP):    else RGroup=3;
573
574
575     * proc print data=distfile n;
576 run;

```

NOTE: Missing values were generated as a result of performing an operation on missing values.

Each place is given by: (Number of times) at (Line):(Column).

3164 at 535:27 743 at 539:27 3164 at 543:27 743 at 560:27

NOTE: There were 3164 observations read from the data set WORK.FINAL.

NOTE: There were 410 observations read from the data set WORK.TOTAL1.

NOTE: The data set WORK.DISTFILE has 3164 observations and 29 variables.

NOTE: DATA statement used (Total process time):

real time 0.05 seconds

cpu time 0.03 seconds

577

```
578 *-----;
579 * Combine schedule K and L to produce K and L report ;
580 * (report 13). Schedule K is report 7 and schedule L is ;
581 * summary of the mixed mail distribution result (finout). ;
582 *-----;
```

583

```
584 proc sort data=distfile;
585     by rgroup route actv;
586     run;
```

NOTE: There were 3164 observations read from the data set WORK.DISTFILE.

NOTE: The data set WORK.DISTFILE has 3164 observations and 29 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time 0.01 seconds

cpu time 0.00 seconds

587

```
588 proc summary data=distfile noprint;
589     var dist_0 dist_1 dist_2 dist_3 dist_4;
590     output out=finout1 sum=dollar0 dollar1 dollar2 dollar3
dollar4;
591     by rgroup route actv;
592     run;
```

NOTE: There were 3164 observations read from the data set WORK.DISTFILE.

NOTE: The data set WORK.FINOUTL has 241 observations and 10 variables.

NOTE: PROCEDURE SUMMARY used (Total process time):

real time 0.01 seconds

cpu time 0.00 seconds

593

```
594 *-----;
595 * The following proc format classifies activity codes into ;
596 * categories for subsequent reports. ;
597 *-----;
598 proc format ;
599 value $actv
600 /*SPECIAL SERVICES */
601     '0060'      ='055 Registered '
```

```

602          '0050'          ='051 Certified   '
603          '0080'          ='054 Insured     '
604          '0030'          ='052 COD         '
605
606 /* OTHSERV INCL. BUSRPLY, RETRCPT, ADDRRCORR, FORMS3547&3579 */
607          '6020','6030'='074 PO Box       '
608          '0090','0190',
609          '0300','0100',
610          '0110','0140'          ='058 Other Services'
611
612 /* FIRST CLASS      */
613          '1060'          ='003 FC SP Letters '
614          '1020'          ='004 FC SP Cards   '
615          '1080','1081','1085','1086'='008 FC Prst Letters'
616          '1022','1035','1040','1045'='009 FC Prst Cards '
617          '2060'          ='014 FC SP Flats   '
618          '2080','2081','2086'          ='017 FC Prst Flats '
619          '3060','4060'          ='143 FC Package Svc '
620          /*'3080','4080'          ='020 FC Prst Parcels'*/
621          '3080','4080'          ='143 FC Package Svc '
622
623
624 /*PRIORITY MAIL    */
625          '1160','2160','3160','4160'='148 Priority       '
626
627 /* EXPRESS MAIL     */
628          '1110','2110','3110','4110' ='140 Express      '
629
630 /* PERIODICALS      */
631          '1211','2211','3211','4211'='031 PER In County   '
632          '1212','2212','3212','4212'='032 PER Outside County'
633
634 /* STANDARD         */
635          '1317'          ='021 MM ECR HD Letters
636          '2317','3317','4317'          ='022 MM ECR HD Flats/Parcels
637          '1318'          ='021 MM ECR SAT Letters
638          '2318','3318','4318'          ='022 MM ECR SAT
639          Flats/Parcels ' '1312','2312','3312','4312'='023 MM ECR Carrier Route
640          '1320','2320','3320','4320'='024 MM EDDM - Retail
641          '1340','1341','1345'          ='025 MM REG Letters
642          '2340','2341','2345'          ='026 MM REG Flats
643          '3340','4340'          ='027 MM REG Parcel
644          '3341','4341'          ='027 MM REG Parcel

```



```

645          '1315'                                ='021 DAL Ltr WSH
646          '2315','3315','4315'                  ='021 DAL Flt/IPP/PAR WSH
647          '1316'                                ='022 DAL Ltr WSS
648          '2316','3316','4316'                  ='022 DAL Flt/IPP/PAR WSS
649          '1319','2319','3319','4319'='021 Parent No Dal HD/SAT
650
651  /* PACKAGE                                */
652          '2402','3402','4402'                    ='157 Ground Advantage HW 1-
31b '
653          '2403','3403','4403'                    ='158 Ground Advantage HW
over31b '
654          '2405','3405','4405'                    ='156 Ground Advantage LW
655          '2410','3410','4410'                    ='145 PKG Retail Ground
656          '2460','2465','2480','2495' ='042 PKG BPM Flats
657          '3460','3465','3480','3495',
658          '4460','4465','4480','4495' ='043 PKG BPM Parcels
659          '1420','1425','1430',
660          '2420','2425','2430',
661          '3420','3425','3430',
662          '4420','4425','4430'                    ='044 PKG Media
663
664  /* PARCEL SELECT and PS Lightweight          */
665          '2360','3360','4360',
666          '2493','3493','4493'                    ='151 PKG Parcel Select
667
668  /* PARCEL RETURN SERVICE (PRS)              */
669          '2475','3475','4475'                    ='154 PKG Parcel Return (PRS)
670
671  /* PREMIUM FORWARDING SERVICE (PFS)          */
672          '0350','1170','2170','3170','4170'='170 Prem. Forw. Svc'
673
674
675  /* GOVERNMENT MAIL                          */
676          '0560',
677          '1510','2510','3510','4510'='085 USPS
678
679  /* FREE                                      */
680          '1910','2910','3910','4910'='086 Free
681
682  /* INTERNATIONAL OUTBOUND AND INBOUND        */
683          '1780','2780','3780','4780',
684          '5081','6081','5082','6082',

```

```

685          '0752','0757','0762','0767',
686          '0770','0860','0862','0867'          ='185

```

International'

687

```

688      /* OTHER CODES */

```

```

689          OTHER = '999 OTHER';

```

NOTE: Format \$ACTV is already on the library WORK.FORMATS.

NOTE: Format \$ACTV has been output.

```

690  run;

```

NOTE: PROCEDURE FORMAT used (Total process time):

real time 0.04 seconds

cpu time 0.01 seconds

691

```

692  data KL (drop=_type_ _freq_);

```

```

693      set rpt7 (in=a) finout1 (in=b);

```

```

694      if a then source='K';

```

```

695      if b then source='L';

```

```

696      if dollar0 > 0 then

```

```

697          do; source='U';

```

```

698              dollar4=dollar0; *move the undistributed mixed

```

mail

```

699                      cost to basic function OTHER;

```

```

700          end;

```

701

```

702      if dollar0=. and dollar1=. and dollar2=. and dollar3=. and

```

```

703          dollar4=. then delete;

```

704

```

705      total= sum (of dollar1 - dollar4);

```

```

706          if actv <= '0900' and actv not in ('0350','0560')

```

```

then actgrp=1;

```

```

707          else if '1020' <= actv <= '4910' or actv in
('0350','0560') then actgrp=2;

```

```

708          else if '5300' <= actv <= '5750' then actgrp=3;

```

```

709          else if '5020' <= actv <= '6720' then actgrp=4; **bls;

```

```

710      **use format to classify activity to classes**;

```

```

711      class=put(actv,$actv.);

```

712

```

713      **assign labels for use in reports**;

```

```

714      label

```

```

715          dollar1 = 'OUTGOING'

```

```

716          dollar2 = 'INCOMING'

```

```

717          dollar3 = 'TRANSIT '

```

```

718          dollar4 = 'OTHER   ';

```

719

```

720          **restrict to letter routes only**;

```

```

721          *if rgroup=1;

```

```

722  run;

```

NOTE: There were 263 observations read from the data set WORK.RPT7.

NOTE: There were 241 observations read from the data set WORK.FINOUTL.

NOTE: The data set WORK.KL has 501 observations and 12 variables.

NOTE: DATA statement used (Total process time):
real time 0.01 seconds
cpu time 0.01 seconds

```
723
724
725 *Generate class variable from lookup table;
726 *import craft,activity code combinations used in delivery cost
model;
727 PROC IMPORT OUT= WORK.ActivityCodesTable
728 DATAFILE=
"&pathprog\ActivityCodesMaster_ForIndicia.xlsx"
SYMBOLGEN: Macro variable PATHPROG resolves to D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles
729 DBMS=XLSX REPLACE;
730 RUN;
```

NOTE: One or more variables were converted because the data type is not supported by the V9

engine. For more details, run with options MSGLEVEL=I.

NOTE: The import data set has 157 observations and 3 variables.

NOTE: WORK.ACTIVITYCODESTABLE data set was successfully created.

NOTE: PROCEDURE IMPORT used (Total process time):

real time 0.01 seconds
cpu time 0.00 seconds

```
731 proc sql;
732 create table KL2 as
733 select KL.*, ac.NewCRAProd as class2
734 from KL left join ActivityCodesTable as ac
735 on KL.actv = ac.ActivityCode;
NOTE: Table WORK.KL2 created, with 501 rows and 13 columns.
```

```
736 quit;
NOTE: PROCEDURE SQL used (Total process time):
real time 0.01 seconds
cpu time 0.00 seconds
```

```
737
738 data checkClass; set KL2;
739 where class ne class2;
740 run;
```

NOTE: There were 219 observations read from the data set WORK.KL2.

WHERE class not = class2;

NOTE: The data set WORK.CHECKCLASS has 219 observations and 13 variables.

NOTE: DATA statement used (Total process time):

real time 0.01 seconds
cpu time 0.01 seconds

```

741
742 ***zzz;
743
744 data KL; set KL2;
745 if class2 = "" then class2 = "999 OTHER";
746 if substr(class2,1,3) = "185" then class = "185 International";
747     else class = class2;
748 drop class2;
749 run;

```

NOTE: There were 501 observations read from the data set WORK.KL2.

NOTE: The data set WORK.KL has 501 observations and 12 variables.

NOTE: DATA statement used (Total process time):

real time	0.01 seconds
cpu time	0.00 seconds

```

750
751 *Export KL table by route, for delivery cost model;
752 proc export data=kl
753     outfile = "&pathOut\KLTable_FCSP_Indicia_Casing&FYData..xlsx"
SYMBOLGEN: Macro variable PATHOUT resolves to D:\ExternalSSD\Folder
19\FY25\SAS\KLTables
SYMBOLGEN: Macro variable FYDATA resolves to 25
754     dbms = xlsx replace;
755 run;

```

NOTE: The export data set has 501 observations and 12 variables.

NOTE: "D:\ExternalSSD\Folder

19\FY25\SAS\KLTables\KLTable_FCSP_Indicia_Casing25.xlsx" file was
successfully created.

NOTE: PROCEDURE EXPORT used (Total process time):

real time	0.04 seconds
cpu time	0.00 seconds